

CLINICAL DOMAIN SPECIFICATION

Cardiovascular Domain

Classification stages, risk calculators, lipid and apolipoprotein logic, safety escalation, and report outputs for SQUARE

PLATFORM ROLE

SQUARE converts uploaded screening results and user context into understandable risk information and suggested next steps. It is not a diagnosis engine and must not replace clinician assessment.

Document field	Specification
Platform	SQUARE - Smart Analysis. Vital Insights. Relevant Action.
Primary setting	Malaysia; adults aged 18 years and above
Domain objective	Upload -> Understand -> Act -> Follow Up
Version	1.0 - cardiovascular domain adjustment
Prepared	1 September 2026

Clinical governance note

All thresholds must be version-controlled against the guideline adopted by SQUARE. Laboratory-specific reference intervals, assay method, units, fasting status, pregnancy status and known disease must be preserved. Outputs should state what was calculated, why it matters, what limitations apply, and when professional review is appropriate.

1. Domain architecture

RECOMMENDED HIERARCHY

Clinical decisions should be driven first by established disease, red flags and overall cardiovascular risk; then by LDL-C/non-HDL-C/ApoB burden. AIP and Castelli ratios provide supplementary pattern explanation and must not independently determine treatment.

Layer	Components	Primary output
Safety	Symptoms; extreme BP; severe hypertriglyceridaemia; acute illness	Emergency, urgent, prompt or routine pathway
Established disease	Prior MI, angina, stroke/TIA, PAD, revascularisation, heart failure	Secondary-prevention pathway; do not rely on FRS
Whole-person risk	Framingham General CVD Risk Score	10-year risk percentage and category
Haemodynamic	Clinic/home BP; treatment status	BP classification and confirmation status
Atherogenic burden	LDL-C, non-HDL-C, ApoB	Primary lipid risk interpretation
Inherited modifier	Lp(a); premature family history	Risk-enhancing flag
Metabolic pattern	TG, HDL-C, glucose, waist, metabolic syndrome	Insulin-resistant/atherogenic pattern
Supplementary indices	AIP; Castelli I and II; ApoB/ApoA-I	Contextual explanation only
Longitudinal	Repeat BP, lipids, weight and actions	Trend and follow-up status

Minimum inputs

- Age, sex at birth, smoking status and diabetes status.
- Systolic and diastolic BP, number of readings, measurement setting and antihypertensive treatment.
- Total cholesterol, LDL-C, HDL-C, triglycerides, units, fasting status and collection date.
- Known ASCVD, CKD, familial hypercholesterolaemia, pregnancy, medications and premature family history.
- Optional advanced tests: ApoB, ApoA-I and Lp(a), retaining laboratory units and reference intervals.

Standard output contract

1. Result and unit
2. Classification or risk category
3. Criteria met and missing inputs
4. Plain-language explanation
5. Important limitations
6. Urgency level and suggested next step
7. Source guideline/version and calculation version

2. Blood-pressure classification

Use the higher category when systolic and diastolic readings fall into different categories. A single measurement is a screening result, not confirmed hypertension.

Classification	SBP (mmHg)	DBP (mmHg)	SQUARE label
Optimal	<120	<80	Optimal
Normal	120-129	80-84	Within normal range
At risk / high-normal	130-139	85-89	Above optimal; monitor
Grade 1 hypertension	140-159	90-99	Raised - confirmation required
Grade 2 hypertension	160-179	100-109	High - prompt clinical review
Grade 3 hypertension	≥ 180	≥ 110	Very high - urgent assessment
Isolated systolic hypertension	≥ 140	<90	Raised systolic BP

Primary local source: Ministry of Health Malaysia, Management of Hypertension, 5th edition (2018).

Confirmation and escalation logic

- Store individual readings; do not average away a single extreme value before safety screening.
- Ask about measurement technique, rest period, cuff size, caffeine/exercise, pain and home versus clinic setting.
- BP $\geq 180/120$ mmHg activates an immediate symptom screen.
- BP $\geq 180/120$ with chest pain, breathlessness, focal weakness, speech difficulty, confusion, severe headache or visual disturbance activates emergency guidance.
- Pregnancy or postpartum status routes to a separate obstetric hypertension pathway.

3. Framingham General Cardiovascular Risk Score

SQUARE should implement the sex-specific 2008 general CVD model used in Malaysian risk assessment, not mix coefficients from the older coronary-only score.

Required input	Validation rule
Sex	Use model-defined male/female equation; document inclusivity limitation
Age	Use only within the adopted model range
Total cholesterol and HDL-C	Same collection date and compatible units
Systolic BP	Pair with treated versus untreated status
Smoking	Current smoking definition must be explicit
Diabetes	Use confirmed status, not one unconfirmed elevated glucose
Established CVD	If present, bypass FRS and enter secondary-prevention pathway

10-year predicted risk	SQUARE category	Display
<10%	Low	Percentage plus risk-factor summary
10-19.9%	Intermediate	Percentage plus risk modifiers
$\geq 20\%$	High	Percentage plus clinician-review recommendation

DO NOT CALCULATE

Do not use FRS as reassurance in established ASCVD, acute symptoms, pregnancy, missing core inputs or populations outside the adopted equation range. CKD, very high LDL-C, familial hypercholesterolaemia and markedly elevated Lp(a) can confer important risk not fully represented by the percentage.

4. Lipid profile and derived indicators

The report should keep measured values visible and separate three concepts: cholesterol concentration, atherogenic particle burden, and whole-person event risk.

4.1 Core measurements

Marker	Primary role	Key implementation note
Total cholesterol	Overall cholesterol concentration	Not a treatment target by itself
LDL-C	Primary cholesterol treatment target	Flag calculated versus directly measured
HDL-C	Risk-pattern component	High HDL-C must not cancel high LDL-C/ApoB
Triglycerides	Metabolic/remnant pattern; pancreatitis safety	Retain fasting status and check extreme values
Non-HDL-C	Cholesterol in all ApoB-containing particles	Useful with high TG, diabetes and metabolic syndrome

$$\text{Non-HDL-C} = \text{Total cholesterol} - \text{HDL-C}$$

4.2 LDL-C calculation

$$\text{Friedewald LDL-C (mmol/L)} = \text{TC} - \text{HDL-C} - \text{TG}/2.2$$

- Only calculate when all components use mmol/L.
- Do not use the Friedewald result when triglycerides are ≥ 4.5 mmol/L or when the laboratory flags the calculation as invalid.
- Prefer the laboratory-reported direct LDL-C or an adopted validated alternative equation when applicable.
- Calculated LDL-C must be labelled as calculated.

4.3 Atherogenic Index of Plasma (AIP)

$$\text{AIP} = \log_{10} [\text{TG} / \text{HDL-C}] \quad (\text{TG and HDL-C in mmol/L})$$

AIP	Conventional research interpretation	Platform status
<0.11	Lower atherogenic pattern	Supplementary
0.11-0.21	Intermediate pattern	Supplementary
>0.21	Higher atherogenic pattern	Supplementary

AIP SAFEGUARD

These bands are commonly used research conventions, not universal treatment thresholds. Prefer a fasting sample. AIP must not override LDL-C, non-HDL-C, ApoB, Lp(a), established disease or overall risk.

4.4 Castelli indices and atherogenic coefficient

$$\text{Castelli I} = \text{TC} / \text{HDL-C} \quad | \quad \text{Castelli II} = \text{LDL-C} / \text{HDL-C}$$

Index	Conventional desirable range	Higher pattern	Important caveat
Castelli I	Men <4.0; women <3.5	Men >5.0; women >4.5	No universal treatment threshold
Castelli II	Men <3.0; women <2.5	At or above desirable boundary	Invalid if LDL-C is unreliable
Atherogenic coefficient	(TC - HDL-C) / HDL-C	Higher is less favourable	Mathematically equals Castelli I - 1

AIP and Castelli outputs are secondary explanatory indices. Clinical guidelines prioritise LDL-C, non-HDL-C, ApoB and total cardiovascular risk.

5. Apolipoproteins and inherited lipid risk

ApoB and Lp(a) should be promoted above ratio-only indices because they add clinically meaningful particle and inherited-risk information.

5.1 Apolipoprotein B

Each major atherogenic particle carries one ApoB molecule. ApoB therefore approximates the number of LDL, IDL, VLDL-remnant and Lp(a) particles rather than the cholesterol mass inside them.

Risk category	Suggested ApoB goal	How SQUARE should present it
Moderate risk	<100 mg/dL	Secondary target linked to risk category
High risk	<80 mg/dL	Above/below risk-linked goal
Very-high risk	<65 mg/dL	Above/below risk-linked goal
Selected recurrent/extreme risk	<55 mg/dL may be considered	Clinician-directed target only

Risk-linked ApoB goals align with ESC/EAS secondary-target conventions; local clinical governance must approve the adopted target set.

Recommended triggers for ApoB

- Elevated triglycerides, diabetes, metabolic syndrome or obesity.
- LDL-C appears acceptable but residual risk is suspected.
- Discordance between LDL-C, non-HDL-C and clinical phenotype.
- Monitoring intensive lipid-lowering therapy when particle burden matters.

5.2 ApoA-I and ApoB/ApoA-I

$$\text{ApoB/ApoA-I ratio} = \text{ApoB} / \text{ApoA-I} \quad (\text{same units})$$

- ApoA-I is the principal structural protein of HDL.

- Use the performing laboratory reference interval for ApoA-I.
- A lower ApoB/ApoA-I ratio indicates a more favourable particle balance; a higher ratio indicates a more atherogenic balance.
- There is no single universally accepted treatment cut-off. Display the numerical ratio and laboratory interpretation, if provided.

5.3 Lipoprotein(a)

Lp(a)	Interpretation
<75 nmol/L	Generally lower
75-124 nmol/L	Intermediate
>=125 nmol/L	Elevated / risk-enhancing
<30 mg/dL	Generally lower
30-49 mg/dL	Intermediate
>=50 mg/dL	Elevated / risk-enhancing

LP(A) UNIT SAFEGUARD

Never apply a fixed conversion between mg/dL and nmol/L. Retain the laboratory unit because particle size makes conversion variable. Lp(a) is largely inherited and is usually measured at least once in adulthood; elevated Lp(a) modifies risk but does not by itself diagnose cardiovascular disease.

6. Lipid interpretation hierarchy and discordance

Pattern	Likely interpretation	Report action
LDL-C high; ApoB high	Concordant high atherogenic burden	Emphasise overall risk and clinician review
LDL-C acceptable; ApoB high	More atherogenic particles than LDL-C suggests	Flag particle discordance
LDL-C high; ApoB relatively lower	Fewer cholesterol-rich particles may be present	Do not dismiss LDL-C; show discordance
TG high; HDL-C low; AIP high	Atherogenic/metabolic pattern	Check glucose, waist, BP and metabolic syndrome
LDL-C controlled; Lp(a) high	Inherited residual-risk modifier	Do not call lipid profile fully reassuring
Ratios favourable; LDL-C/ApoB high	Ratios may obscure absolute burden	Prioritise absolute atherogenic measures

Recommended interpretation sequence

8. Check safety and established disease.
9. Validate sample, units, fasting status and calculation eligibility.
10. Classify LDL-C, triglycerides and HDL-C; calculate non-HDL-C.
11. Apply overall cardiovascular risk and risk-linked targets.
12. Interpret ApoB and check discordance.
13. Apply Lp(a) and family history as risk modifiers.
14. Display AIP and Castelli ratios as supplementary pattern indicators.

15. Generate action and follow-up language without diagnosing.

Familial hypercholesterolaemia trigger

POSSIBLE FH

Markedly elevated LDL-C (commonly around ≥ 4.9 mmol/L in adults), premature ASCVD, tendon xanthomata or a strong family history should trigger a validated familial-hypercholesterolaemia pathway and clinician assessment. A single lipid value should not automatically diagnose FH.

Hypertriglyceridaemia safety

Triglycerides	Platform action
<1.7 mmol/L	Within desirable range
1.7-4.4 mmol/L	Raised; assess metabolic and secondary causes
≥ 4.5 mmol/L	Do not use Friedewald LDL-C; prompt clinical review
≥ 10 mmol/L	Severe; urgent clinical assessment because pancreatitis risk rises substantially

7. Metabolic and kidney risk integration

Cardiovascular interpretation should reuse validated outputs from other domains rather than recalculate them inconsistently.

7.1 Metabolic syndrome

Criteria are met when any three of the following five components are present:

Component	Positive criterion
Central obesity	Asian men waist ≥ 90 cm; Asian women ≥ 80 cm
Triglycerides	≥ 1.7 mmol/L or treatment
HDL-C	<1.0 mmol/L in men; <1.3 mmol/L in women; or treatment
Blood pressure	$\geq 130/85$ mmHg or antihypertensive treatment
Fasting glucose	≥ 5.6 mmol/L or known diabetes/treatment

If any component is missing, display the number met and “cannot fully determine” when the missing value could change the result.

7.2 Kidney risk

Renal marker	Cardiovascular relevance
eGFR <60 mL/min/1.73 m ²	Potential CKD risk enhancer if persistent for >3 months
Albuminuria A2 or A3	Independent cardiovascular and kidney-risk marker
G4-G5 CKD	High/very-high clinical risk; FRS may underestimate

Single abnormal result	Requires chronicity or other kidney-damage evidence before CKD label
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8. Safety and escalation matrix

Trigger	Urgency	Minimum platform response
Chest pain, acute breathlessness, stroke symptoms, collapse	Emergency	Advise immediate emergency services; stop routine scoring
BP $\geq$ 180/120 with concerning symptoms	Emergency	Emergency pathway
BP $\geq$ 180/120 without concerning symptoms	Urgent	Repeat correctly; urgent same-day assessment guidance
Triglycerides $\geq$ 10 mmol/L	Urgent	Prompt clinical assessment; symptom screen for pancreatitis
Marked LDL-C with premature personal/family CVD	Prompt	Possible FH pathway and clinician referral
New high FRS or high ApoB/Lp(a) modifier	Routine to prompt	Clinician cardiovascular-risk review
Single mildly abnormal ratio only	Routine	Lifestyle context and repeat with core markers

9. Report-generation rules

Each cardiovascular report should produce the following blocks in this order.

Block	Required content
Safety status	No red flag / prompt review / urgent / emergency
Headline	One sentence combining BP, lipid burden and whole-person risk
BP	Readings, setting, category, confirmation status
10-year risk	FRS percentage/category or reason not calculated
Core lipids	TC, LDL-C, HDL-C, TG and non-HDL-C with units
Advanced lipids	ApoB, ApoA-I, ApoB/ApoA-I and Lp(a), when available
Supplementary indices	AIP and Castelli I/II with limitations
Risk modifiers	Diabetes, CKD, smoking, family history, waist and Lp(a)
Why this result	Criteria met and discordance explanation
Actions	Emergency/referral/lifestyle/repeat-test guidance
Follow-up	Suggested monitoring items and clinician-confirmed interval

Example output language

BALANCED LIPID EXPLANATION

Your LDL cholesterol is within the stated laboratory range, but ApoB is above the goal associated with your risk category. This means the number of cholesterol-carrying particles may be higher than LDL cholesterol alone suggests. The AIP and cholesterol ratios support the same overall pattern, but they are supplementary indicators rather than treatment targets.

RISK-SCORE EXPLANATION

Your Framingham result estimates the chance of a major cardiovascular event over the next 10 years using age, sex, cholesterol, blood pressure, smoking and diabetes information. It is a population-based estimate, not a prediction of what will definitely happen to you.

Prohibited claims

- “You have blocked arteries” based on AIP or a Castelli ratio.
- “No cardiovascular risk” because HDL-C is high or FRS is low.
- “You have familial hypercholesterolaemia” from one LDL-C result.
- “Your heart is healthy” from a normal lipid profile.
- Medication initiation, cessation or dose changes without clinician involvement.

10. Quality assurance and calculator validation

- Unit tests for mmol/L and mg/dL conversion.
- Golden-case comparison against the adopted FRS reference implementation.
- Boundary tests immediately below, at and above every classification threshold.
- Missing-value and contradictory-value tests.
- Sex-, age-, treatment- and disease-exclusion tests.
- Independent clinical review of all report phrases and red-flag routes.
- Versioned coefficients, thresholds, references and release dates.
- Audit log preserving source values, transformed values, formula version and output.

11. Implementation-ready calculator register

ID	Calculator / classification	Inputs	Output priority
CV-01	BP classification	SBP, DBP, setting, readings	Core
CV-02	Framingham General CVD	Age, sex, TC, HDL-C, SBP, treatment, smoking, diabetes	Core
CV-03	Non-HDL-C	TC, HDL-C	Core
CV-04	Calculated LDL-C	TC, HDL-C, TG	Core with validity gate
CV-05	AIP	TG, HDL-C in mmol/L	Supplementary
CV-06	Castelli I	TC, HDL-C	Supplementary
CV-07	Castelli II	LDL-C, HDL-C	Supplementary
CV-08	Atherogenic coefficient	TC, HDL-C	Optional / hidden

CV-09	ApoB target	ApoB, global risk category	Advanced core
CV-10	ApoB/ApoA-I	ApoB, ApoA-I	Supplementary
CV-11	Lp(a) modifier	Lp(a), native unit	Advanced core
CV-12	Metabolic syndrome	Waist, TG, HDL-C, BP, glucose/treatment	Integrated
CV-13	FH trigger	LDL-C, personal/family history, signs	Referral trigger
CV-14	TG safety	TG, symptoms	Safety

12. References and governance sources

- Ministry of Health Malaysia. Management of Hypertension, 5th edition. 2018.
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- KDIGO. Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. 2024. [https://www.kidney-international.org/article/S0085-2538\(23\)00766-4/fulltext](https://www.kidney-international.org/article/S0085-2538(23)00766-4/fulltext)
- SQUARE clinical governance requirement: validate all adopted thresholds against current Malaysian guidance before production release.

Approval checkpoint

BEFORE PRODUCTION

A Malaysian-registered medical practitioner should approve the exact FRS equation, lipid targets, ApoB goals, Lp(a) bands, AIP/Castelli wording, emergency routes and version-control process. Product copy must preserve SQUARE's non-diagnostic role.